SIDDHARTH SHARMA

B.Tech, Computer Science and Engineering (IoT), Manipal University Jaipur · Expected 2028

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SUMMARY

Engineering student who builds automated trading systems and full-stack products end to end. I run my own live multi-strategy trading stack on a self-managed Linux server, and I ship web and mobile apps in Next.js, React Native and Python.

EDUCATION

Manipal University Jaipur 2024 – 2028

B.Tech in Computer Science and Engineering (Internet of Things)

PROJECTS

Stellar Capital — Algorithmic Trading Platform *Python, Linux VPS, systemd, broker APIs*

- A single control panel over all my trading accounts (Groww, Delta Exchange). It monitors the live bots, enforces risk limits, keeps a trade journal, sends a weekly review PDF and includes an AI coach.
- Runs several live strategy books across NSE equities, crypto and gold perpetual futures, on one VPS with a shared broker rate limiter and clean-log checks.
- Every strategy change is backtested before deployment on point-in-time NSE data from 2014 to 2026, including delisted stocks, with fees, slippage and split-period checks.

India Whales — Unusual Options Activity Tracker for NSE *Python, live tick feed, web app*

- Detects bursts from a single large buyer in real time across about 750 near-expiry stock option contracts, using the Groww tick feed.
- Adds bulk and block deals, SEBI insider buys (parsed from XBRL), short sales and delivery spikes. Each alert is graded five days later. Public website coming soon.

Joytish IQ — Vedic Astrology Engine and Dashboard *Python, Skyfield, Next.js 16, Recharts*

- Computes planet positions from the NASA JPL DE421 ephemeris, with a hand-built Lahiri ayanamsa that matches PyEphem within 22 arc-seconds. Also calculates dashas, yogas, ashtakavarga and a life timeline.
- The dashboard shows an outlook, life periods and a planner. A public version, where anyone can enter their date of birth, is coming soon.

Poolmate — Ride-Pooling Platform *Next.js 15, React, Supabase, Leaflet, Expo*

- A rider web app with seat and full-cab booking, live maps, in-app chat, support tickets and an admin panel, plus a React Native (Expo) driver app.

Options Manager — Index Options Analyser *Python*

- A Black-Scholes pricing engine for Nifty and Bank Nifty options, with a feasibility check and a ranking of option strategies by risk and reward.

RESEARCH

Momentum in Indian Equities, 1997–2026 *Student paper, MUJ · SSRN submission planned*

- Tested "buy last year's winners" on NSE with point-in-time data. A low-drawdown variant (the top 10 Nifty 200 stocks near their 52-week high) returned 24.4% CAGR with a maximum drawdown of −28%.
- Classic 12-1 momentum returned about 23% a year after brokerage fees and tax. Simple trend filters cut the crash drawdown roughly in half.

Hybrid CNN–Transformer for Lung Cancer Histopathology *Academic project (PBL)*

- Designed a ConvNeXt + DeiT fusion model, to be tested on patient-level splits (LungHist700) that avoid the data leakage in image-level results reported on LC25000.

IN PROGRESS

- **Study AI** — An AI tutor that teaches from uploaded course files, with voice, interruption and resume. *Next.js 16, SQLite FTS5, LLM provider adapters.*
- **Tars** — An AI-integrated hardware desk robot.
- **PropertyDheko** — A housing marketplace for buyers, sellers and renters. *Next.js, Supabase.*

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Web and mobile: Next.js, React, Tailwind CSS, React Native (Expo), Supabase, SQLite

Data and quant: pandas, NumPy, Numba, backtesting, walk-forward and Monte Carlo testing, Black-Scholes

Infrastructure: Linux VPS, systemd, Git and GitHub, broker APIs (Groww, Delta Exchange), Telegram bots